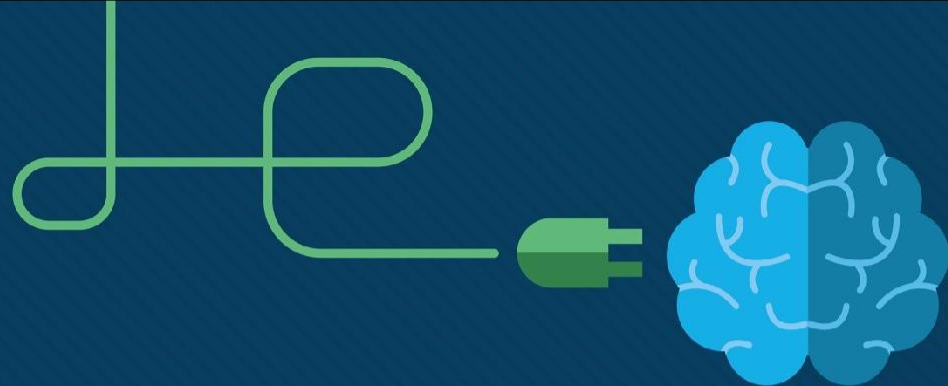




Basic Switch and End Device Configuration

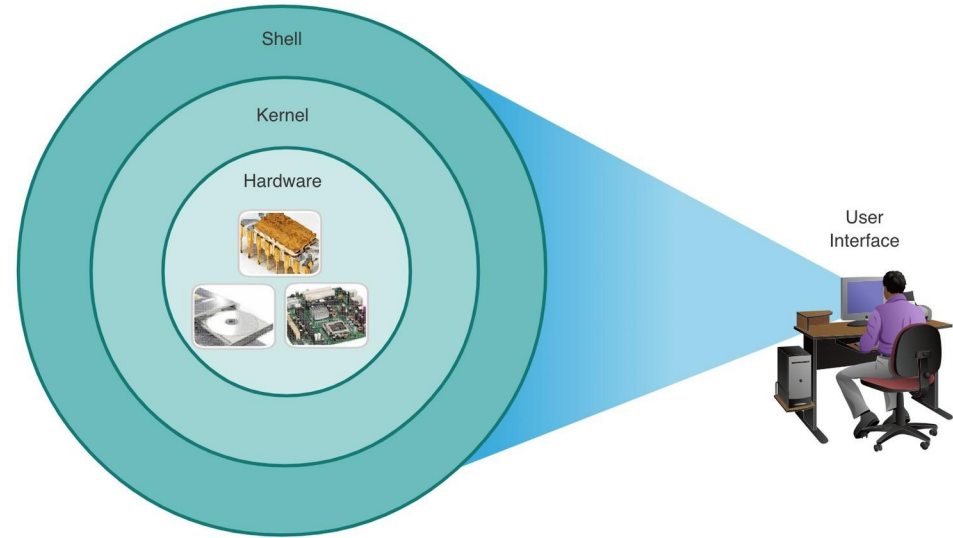
Introductions to Networks v7.0
(ITN)



Cisco IOS Access

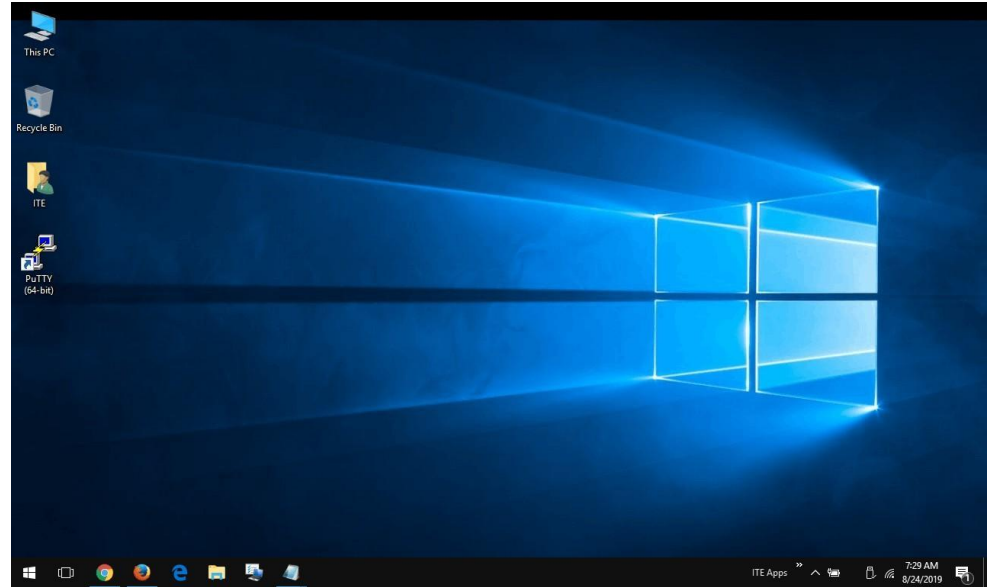
Operating Systems

- **Shell** - The user interface that allows users to request specific tasks from the computer. These requests can be made either through the CLI or GUI interfaces.
- **Kernel** - Communicates between the hardware and software of a computer and manages how hardware resources are used to meet software requirements.
- **Hardware** - The physical part of a computer including underlying electronics.



Cisco IOS Access GUI

- A GUI allows the user to interact with the system using an environment of graphical icons, menus, and windows.
- A GUI is more user-friendly and requires less knowledge of the underlying command structure that controls the system.
- Examples of these are: Windows, macOS, Linux KDE, Apple iOS and Android.
- GUIs can fail, crash, or simply not operate as specified. For these reasons, network devices are typically accessed through a CLI.



Cisco IOS Access

Purpose of an OS

PC operating system enables a user to do the following:

- Use a mouse to make selections and run programs
- Enter text and text-based commands



CLI-based network operating system enables a network technician to do the following:

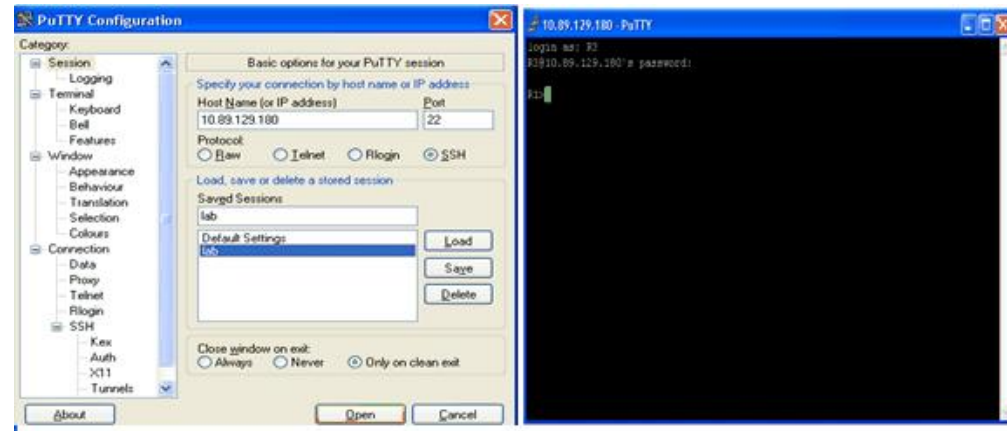
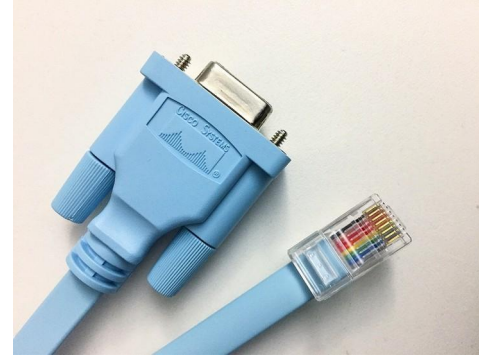
- Use a keyboard to run CLI-based network programs
- Use a keyboard to enter text and text-based commands
- View output on a monitor

```
analyst@secOps ~]$ ls
Desktop  Downloads  lab.support.files  second_drive
[analyst@secOps ~]$
```

Cisco IOS Access

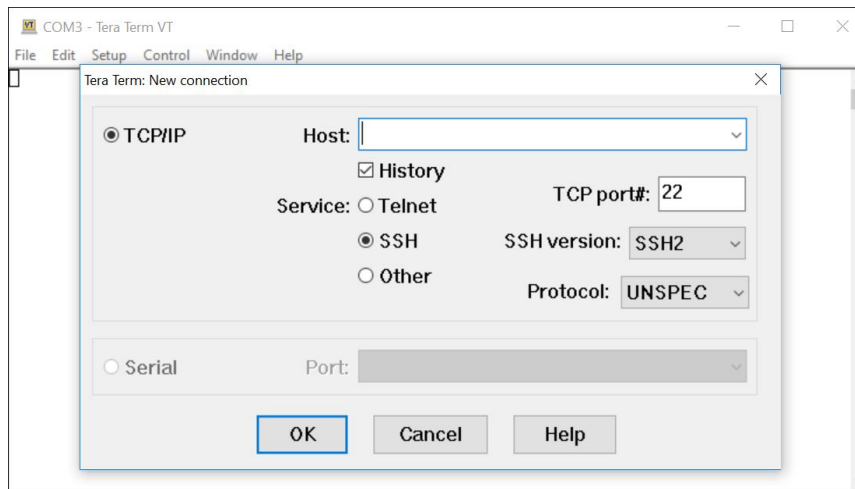
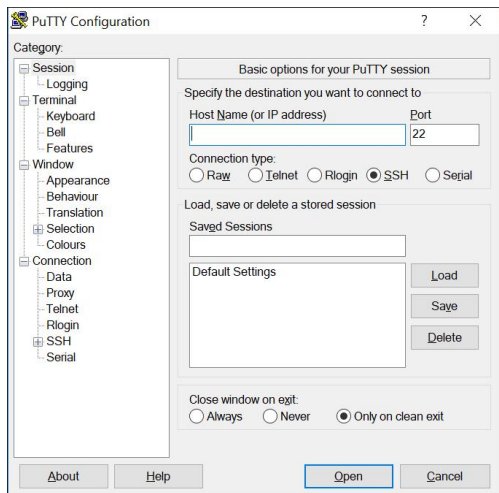
Access Methods

- **Console** – A physical management port used to access a device in order to provide maintenance, such as performing the initial configurations.
- **Secure Shell (SSH)** – Establishes a secure remote CLI connection to a device, through a virtual interface, over a network. (Note: This is the recommended method for remotely connecting to a device.)
- **Telnet** – Establishes an insecure remote CLI connection to a device over the network. (Note: User authentication, passwords and commands are sent over the network in plaintext.)



Terminal Emulation Programs

- Terminal emulation programs are used to connect to a network device by either a console port or by an SSH/Telnet connection.
- There are several terminal emulation programs to choose from such as PuTTY, Tera Term and SecureCRT.



IOS Navigation

Primary Command Modes

User EXEC Mode:

- Allows access to only a limited number of basic monitoring commands
- Identified by the CLI prompt that ends with the > symbol

```
Router>
```

```
Switch>
```

Privileged EXEC Mode:

- Allows access to all commands and features
- Identified by the CLI prompt that ends with the # symbol

```
Router#
```

```
Switch#
```

Configuration Mode and Subconfiguration Modes

Global Configuration Mode:

- Used to access configuration options on the device

```
Switch(config) #
```

Line Configuration Mode:

- Used to configure console, SSH, Telnet or AUX access

```
Switch(config-line) #
```

Interface Configuration Mode:

- Used to configure a switch port or router interface

```
Switch(config-if) #
```

Video – IOS CLI Primary Command Modes

This video will cover the following:

- User EXEC mode
- Privilege EXEC mode
- Global Config mode

Navigation Between IOS Modes

▪ Privileged EXEC Mode:

- To move from user EXEC mode to privilege EXEC mode, use the **enable** command.

```
Switch> enable  
Switch#
```

▪ Global Configuration Mode:

- To move in and out of global configuration mode, use the **configure terminal** command. To return to privilege EXEC mode, use the **exit** command.

```
Switch(config)#  
Switch(config)#exit  
Switch#
```

▪ Line Configuration Mode:

- To move in and out of line configuration mode, use the **line** command followed by the management line type. To return to global configuration mode, use the **exit** command.

```
Switch(config)#line console 0  
Switch(config-line)#exit  
Switch(config)#
```

Navigation Between IOS Modes (Cont.)

Subconfiguration Modes:

- To move out of any subconfiguration mode to get back to global configuration mode, use the **exit** command. To return to privilege EXEC mode, use the **end** command or key combination **Ctrl +Z**.
- To move directly from one subconfiguration mode to another, type in the desired subconfiguration mode command. In the example, the command prompt changes from **(config-line)#** to **(config-if)#**.

```
Switch(config)#line console 0
Switch(config-line)#end
Switch#
```

```
Switch(config-line)#interface FastEthernet 0/1
Switch(config-if)#
```

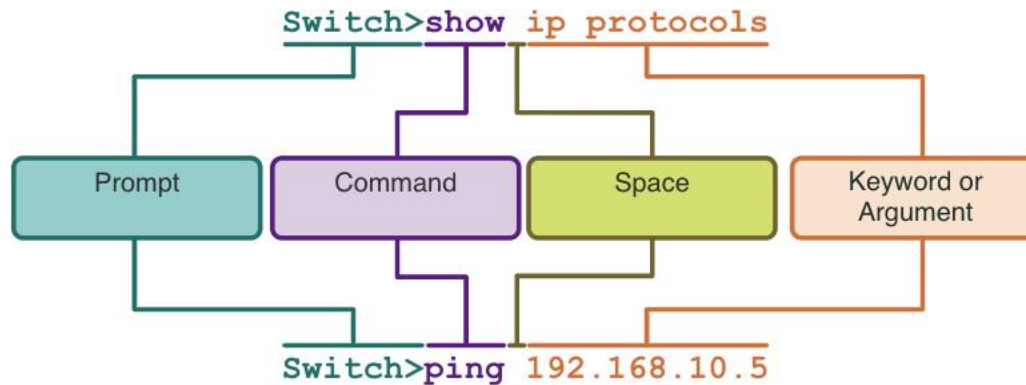
Video – Navigation Between IOS Modes

This video will cover the following:

- enable
- disable
- configure terminal
- exit
- end
- Control + Z on keyboard
- Other commands to enter sub configuration modes

2.3 The Command Structure

Basic IOS Command Structure



- **Keyword** – This is a specific parameter defined in the operating system (in the figure, **ip protocols**).
- **Argument** - This is not predefined; it is a value or variable defined by the user (in the figure, **192.168.10.5**).

2.4 Basic Device Configuration

Device Names

- The first configuration command on any device should be to give it a unique hostname.
- By default, all devices are assigned a factory default name. For example, a Cisco IOS switch is "Switch."
- Guideline for naming devices:
 - Start with a letter
 - Contain no spaces
 - End with a letter or digit
 - Use only letters, digits, and dashes
 - Be less than 64 characters in length

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Note: To return the switch to the default prompt, use the **no hostname** global config command.

Password Guidelines

- The use of weak or easily guessed passwords are a security concern.
- All networking devices should limit administrative access by securing privileged EXEC, user EXEC, and remote Telnet access with passwords. In addition, all passwords should be encrypted and legal notifications provided.
- Password Guidelines:
 - Use passwords that are more than eight characters in length.
 - Use a combination of upper and lowercase letters, numbers, special characters, and/or numeric sequences.
 - Avoid using the same password for all devices.
 - Do not use common words because they are easily guessed.



Note: Most of the labs in this course use simple passwords such as **cisco** or **class**. These passwords are considered weak and easily guessable and should be avoided in production environments.

Basic Device Configuration

Configure Passwords

Securing user EXEC mode access:

- First enter line console configuration mode using the **line console 0** command in global configuration mode.
- Next, specify the user EXEC mode password using the **password** *password* command.
- Finally, enable user EXEC access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Securing privileged EXEC mode access:

- First enter global configuration mode.
- Next, use the **enable secret** *password* command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Configure Passwords (Cont.)

Securing VTY line access:

- First enter line VTY configuration mode using the **line vty 0 15** command in global configuration mode.
- Next, specify the VTY password using the **password *password*** command.
- Finally, enable VTY access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

- Note: VTY lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 VTY lines that are numbered 0 to 15.

Basic Device Configuration

Encrypt Passwords

- The startup-config and running-config files display most passwords in plaintext.
- To encrypt all plaintext passwords, use the **service password-encryption** global config command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

- Use the **show running-config** command to verify that the passwords on the device are now encrypted.

```
Sw-Floor-1# show running-config
!
!
line con 0
password 7 094F471A1A0A
login
!
Line vty 0 4
Password 7 03095A0F034F38435B49150A1819
Login
!
!
end
```

Basic Device Configuration

Banner Messages

- A banner message is important to warn unauthorized personnel from attempting to access the device.
- To create a banner message of the day on a network device, use the **banner motd # the message of the day #** global config command.

Note: The “#” in the command syntax is called the delimiting character. It is entered before and after the message.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only!#
```

The banner will be displayed on attempts to access the device.



```
Press RETURN to get started.
```

```
Authorized Access Only!
```

```
User Access Verification
```

```
Password:
```

2.5 Save Configurations

Save Configurations

Configuration Files

- There are two system files that store the device configuration:
 - **startup-config** - This is the saved configuration file that is stored in NVRAM. It contains all the commands that will be used by the device upon startup or reboot. Flash does not lose its contents when the device is powered off.
 - **running-config** - This is stored in Random Access Memory (RAM). It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately. RAM is volatile memory. It loses all of its content when the device is powered off or restarted.
 - To save changes made to the running configuration to the startup configuration file, use the **copy running-config startup-config** privileged EXEC mode command.

```
Router#show startup-config
Using 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
Router#show running-config
Building configuration...

Current configuration : 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

Alter the Running Configurations

If changes made to the running config do not have the desired effect and the running-config has not yet been saved, you can restore the device to its previous configuration. To do this you can:

- Remove the changed commands individually.
- Reload the device using the **reload** command in privilege EXEC mode. *Note: This will cause the device to briefly go offline, leading to network downtime.*

If the undesired changes were saved to the startup-config, it may be necessary to clear all the configurations using the **erase startup-config** command in privilege EXEC mode.

- After erasing the startup-config, reload the device to clear the running-config file from RAM.

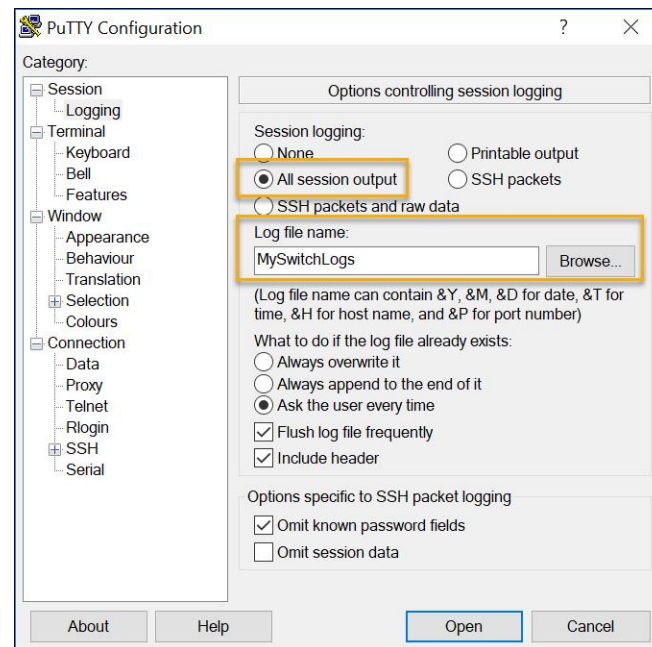
```
Router# reload
Proceed with reload? [confirm]
Initializing Hardware ...
```

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#
```

Capture Configuration to a Text File

Configuration files can also be saved and archived to a text document.

- **Step 1.** Open terminal emulation software, such as PuTTY or Tera Term, that is already connected to a switch.
- **Step 2.** Enable logging in to the terminal software and assign a name and file location to save the log file. The figure displays that **All session output** will be captured to the file specified (i.e., MySwitchLogs).

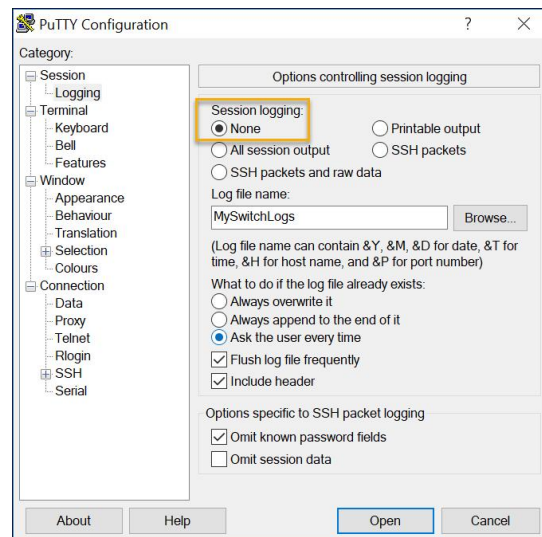


Capture Configuration to a Text File (Cont.)

- **Step 3.** Execute the **show running-config** or **show startup-config** command at the privileged EXEC prompt. Text displayed in the terminal window will be placed into the chosen file.
- **Step 4.** Disable logging in the terminal software. The figure shows how to disable logging by choosing the **None** session logging option

Note: The text file created can be used as a record of how the device is currently implemented. The file could require editing before being used to restore a saved configuration to a device.

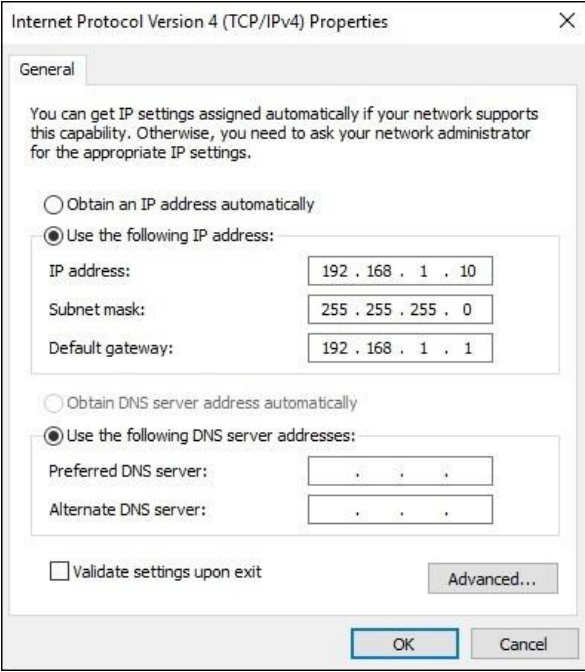
```
Switch# show running-config
Building configuration...
```



2.6 Ports and Addresses

IP Addresses

- The use of IP addresses is the primary means of enabling devices to locate one another and establish end-to-end communication on the internet.
- The structure of an IPv4 address is called dotted decimal notation and is represented by four decimal numbers between 0 and 255.
- An IPv4 subnet mask is a 32-bit value that differentiates the network portion of the address from the host portion. Coupled with the IPv4 address, the subnet mask determines to which subnet the device is a member.
- The default gateway address is the IP address of the router that the host will use to access remote networks, including the internet.



The screenshot shows the 'Internet Protocol Version 4 (TCP/IPv4) Properties' dialog box with the 'General' tab selected. The dialog box contains instructions on how to obtain IP settings and options for manual configuration. The 'Use the following IP address' option is selected, with the IP address set to 192.168.1.10, the subnet mask to 255.255.255.0, and the default gateway to 192.168.1.1. The 'Use the following DNS server addresses' option is also selected, with both preferred and alternate DNS server fields empty. There are buttons for 'OK', 'Cancel', and 'Advanced...'. A checkbox for 'Validate settings upon exit' is present but unchecked.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

IP Addresses (Cont.)

- IPv6 addresses are 128 bits in length and written as a string of hexadecimal values. Every four bits is represented by a single hexadecimal digit; for a total of 32 hexadecimal values. Groups of four hexadecimal digits are separated by a colon “.”.

- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.

Note: IP in this course refers to both the IPv4 and IPv6 protocols. IPv6 is the most recent version of IP and is replacing the more common IPv4.

The screenshot shows the 'Internet Protocol Version 6 (TCP/IPv6) Properties' dialog box with the 'General' tab selected. The dialog box contains instructions on how to obtain IPv6 settings. Two radio buttons are present: 'Obtain an IPv6 address automatically' (unselected) and 'Use the following IPv6 address:' (selected). Below the selected option are three text input fields: 'IPv6 address:' containing '2001:db8:acad:10::10', 'Subnet prefix length:' containing '64', and 'Default gateway:' containing 'fe80::1'. Below these fields are two more radio buttons: 'Obtain DNS server address automatically' (unselected) and 'Use the following DNS server addresses:' (selected). Below the selected option are two text input fields: 'Preferred DNS server:' and 'Alternate DNS server:'. At the bottom left is a checkbox labeled 'Validate settings upon exit' which is unchecked. At the bottom right are three buttons: 'Advanced...', 'OK', and 'Cancel'.

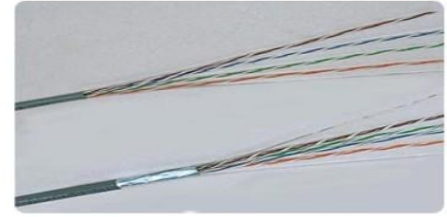
Ports and Addresses

Interfaces and Ports

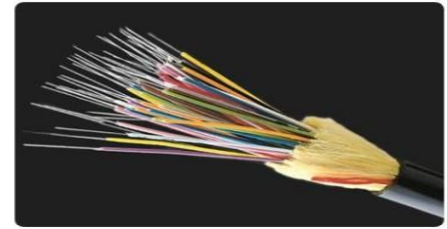
-
-
-



Copper



Fiber-optics



Wireless



2.7 Configure IP Addressing

Manual IP Address Configuration for End Devices

- End devices on the network need an IP address in order to communicate with other devices on the network.
- IPv4 address information can be entered into end devices manually, or automatically using Dynamic Host Configuration Protocol (DHCP).
- To manually configure an IPv4 address on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
- Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window. Then configure the IPv4 address and subnet mask information, and default gateway.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

Note: IPv6 addressing and configuration options are similar to IPv4.

Automatic IP Address Configuration for End Devices

- DHCP enables automatic IPv4 address configuration for every end device that is DHCP-enabled.
- End devices are typically by default using DHCP for automatic IPv4 address configuration.
- To configure DHCP on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
- Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window, then select **Obtain an IP address automatically** and **Obtain DNS server address automatically**.



Note: IPv6 uses DHCPv6 and SLAAC (Stateless Address Autoconfiguration) for dynamic address allocation.

Switch Virtual Interface Configuration

To access the switch remotely, an IP address and a subnet mask must be configured on the SVI.

To configure an SVI on a switch:

- Enter the **interface vlan 1** command in global configuration mode.
- Next assign an IPv4 address using the **ip address** *ip-address subnet-mask* command.
- Finally, enable the virtual interface using the **no shutdown** command.

```
Switch# configure terminal
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.20 255.255.255.0
Switch(config-if)# no shutdown
```

Packet Tracer – Implement Basic Connectivity

In this Packet Tracer, you will do the following:

- Perform a Basic Configuration on two switches
- Configure the PCs
- Configure the Switch Management Interface

